Multi-Label Prediction and Explanation of Drug–Herb–Supplement Adverse Events in Oncology Patients

Feature Selection and Post-Hoc Explainability for Polypharmacy Risk, Using U.S. FAERS Data

Program	Track	Term	Faculty Mentor
NSF REU, Worcester Polytechnic Institute	AI for Social Good	Summer 2026	Dr. Chun-Kit Ngan, WPI

Developed with Amruta Soma (equal contribution), Gandhali Sheth, and Bo (Eric) Wang, under the supervision of Dr. Chun-Kit Ngan (WPI) and Dr. Yin-Ting Cheung (CUHK School of Pharmacy).

Background & Motivation

Adverse events (AEs) — unintended, harmful reactions to a medical intervention — are among the leading causes of morbidity and mortality in modern healthcare, estimated to affect over 2.2 million hospitalized patients and contribute to more than 100,000 deaths annually in the United States alone. Cancer patients carry a disproportionate share of this risk: polypharmacy, typically defined as the concurrent use of five or more medications, affects roughly three in four cancer patients and grows more common with age, and drug-drug interactions alone are estimated to account for around 30% of all reported adverse events.

That picture becomes more complex once complementary therapies enter the regimen. Globally, an estimated 22% of cancer patients report using herbal medicine, with usage climbing to roughly 40% in Africa, 28% in Asia, and 17–20% in North America. Herbal use is especially embedded in Traditional Chinese Medicine (TCM), a healthcare system in which botanical formulations make up the majority of clinical encounters; regional studies report that more than 60% of pediatric cancer patients and over 10% of adult cancer patients turn to TCM to manage side effects of their disease and its treatment. Dietary supplement use follows a similar trajectory in the United States, where 52% of adults report taking at least one supplement in any given month — and among cancer patients specifically, just over half report taking supplements concurrently with active chemotherapy, with roughly 41% using some form of complementary or alternative medicine (most commonly vitamins and multivitamins) during anti-cancer treatment. Unlike herbal remedies, which patients often take for a perceived effect on the cancer itself, supplements are more commonly taken for general wellness and are frequently never disclosed to oncology providers — leaving many drug-supplement interactions undetected until they manifest clinically.

Because drugs, herbs, and supplements each carry distinct mechanisms of interaction and risk, this project treats them as separate exposure categories rather than folding them into one generic label. Most existing computational approaches to adverse event prediction instead rely on chemical and biological docking simulations, which model risk from a compound's molecular structure — a reasonable approach for a single pharmaceutical drug, but one that breaks down for herbal products, whose multicomponent and often uncharacterized chemistry resists the same analysis. These docking-based methods also largely ignore patient-level risk factors (age, sex, comorbidities) and typically treat each adverse event as an independent, single-label outcome, despite clinical evidence that adverse events in oncology patients frequently occur in recognizable, correlated clusters rather than in isolation. Closing that gap — predicting multiple, interacting adverse events from patient-level exposure data, and explaining which factors are driving them — is what motivated this project's two central contributions.

Data & Problem Setup

The project draws on roughly 1.2 million adverse event reports from FAERS (the FDA Adverse Event Reporting System), focused on oncology patients and organized into drug/herb/supplement (DHS) exposure categories. Because herb- and supplement-reporting coverage is comparatively sparse across pharmacovigilance data generally, this phase of evaluation focused on the two most complete cohorts: patients exposed to drugs and herbs, and patients exposed to drugs, herbs, and supplements together. Adverse event prediction is framed as genuine multi-label classification — a patient can be positive for several adverse event categories at once — evaluated across three standard multi-label problem transformations (Binary Relevance, Classifier Chains, Label Powerset) and nine base classifiers spanning tree-based, margin-based, connectionist, and probabilistic model families, for 81 model configurations per exposure cohort.

Contribution 1 — ARISES: Interaction-Aware Feature Selection

ARISES (Association Rule Interaction Scoring and Ensemble Selection) is our feature selection framework, purpose-built for the multi-label, DHS-interaction structure described above. Rather than scoring each feature, or each adverse event, in isolation the way standard filter, wrapper, and embedded feature selection methods do, ARISES first mines the data for two kinds of interaction signal: statistically significant drug/herb/supplement combinations associated with specific adverse events — validated using established pharmacovigilance disproportionality metrics with multiple-comparison correction — and adverse event combinations that reliably co-occur across the patient population, validated by consensus across independent frequent-itemset mining algorithms. These interaction signals are engineered into new composite features, which then compete on equal footing with the original clinical covariates in an ensemble voting selection stage that combines three complementary, established multi-label feature selection methods rather than relying on any single method's assumptions about how labels relate to one another.

HEADLINE RESULT

ARISES beat the strongest of 81 baseline model configurations on **6 of 7** multi-label evaluation metrics, in both oncology exposure cohorts tested.

Drug & Herb cohort	Drug, Herb & Supplement cohort
+4.29% F1	**+2.06% F1**
+4.19% Jaccard	**+3.72% Jaccard**
avg. improvement over the best baseline, per metric	avg. improvement over the best baseline, per metric

Figures reflect the F1 and Jaccard metrics (micro/macro/weighted variants averaged), the two most commonly reported multi-label performance measures, computed against the strongest individual baseline score achieved for each metric rather than a single overall-best baseline model.

What makes it different

- Builds candidate features from two complementary interaction sources: DHS co-prescription patterns validated with population-level statistical disproportionality analysis, and adverse-event co-occurrence patterns validated through consensus across multiple independent itemset-mining algorithms.
- Selects the final feature set through an ensemble vote across several established, methodologically distinct selection strategies, rather than trusting any single method's ranking.
- Directly targets the multi-label structure of the problem, so the selected features reflect how adverse events and exposures interact — not just how each behaves in isolation.

Contribution 2 — MLX: Multi-Label Explainability

A model that predicts several interacting adverse events is only useful to a pharmacovigilance reviewer if they can understand why it made that prediction — and general-purpose explainability tools such as LIME and SHAP (and fast approximations like FastSHAP) were built around a single scalar output, so in a multi-label setting they explain one label at a time and cannot express how one predicted adverse event relates to another. Even the small number of explainability methods purpose-built for multi-label problems either bake interpretability into a specific, non-transferable model architecture, or represent a label combination as nothing more than the union of independently derived single-label explanations. **MLX** (Multi-Label eXplainer) takes a different starting point: the co-occurring group of predicted adverse events, not the individual label, is the unit of explanation.

Given a patient's predicted adverse event set, MLX (1) identifies which statistically meaningful combinations of those events are relevant to this specific patient, using population-level association patterns mined earlier in the pipeline; (2) computes a personalized statistic measuring how strongly the trained model actually couples those exact labels for this individual, as distinct from how often the combination appears in the population at large; and (3) runs a structured, gradient-based counterfactual search to identify which clinical and DHS features are driving that joint prediction, separating the output into actionable recommendations (medications a clinician could reasonably adjust) and non-actionable risk context (e.g., age or baseline health conditions). Every attribution is then checked against real, clinically similar patients in the dataset, so the pipeline reports its own confidence rather than presenting every explanation with equal certainty.

Why it's novel

MLX extends the counterfactual explanation framing established in prior XAI literature — gradient-based search for a minimal perturbation that flips a prediction, and continuous relaxation for searching mixed binary/continuous feature spaces — from a single scalar prediction to the *joint* probability of a label group. That shift is what lets MLX explain why a group of adverse events was predicted *together*, as its own explanatory unit, rather than explaining each label independently and leaving a reviewer to mentally reassemble the interactions.

Prototype

To make the pipeline usable outside a research script, we also built a lightweight, web-based proof-of-concept: a user inputs a hypothetical medication and clinical profile and receives, in return, a predicted adverse event set alongside an interactive, MLX-driven explanation — including which mined adverse event groups behave as correlated for that patient, which DHS combinations the model flags as actionable, and which patient characteristics are contributing risk context. The prototype is a methodological demonstration, not a clinically validated decision-support tool.

Why This Matters

Complementary and alternative medicine use — herbal formulations rooted in traditions like TCM, and dietary supplements increasingly taken alongside cancer treatment in the U.S. — is common, often undisclosed to providers, and correspondingly under-characterized in standard pharmacovigilance workflows. By pairing a feature selection method that surfaces genuine DHS interaction signal (ARISES) with an explainability method that reasons about adverse events jointly instead of one at a time (MLX), this project moves toward decision-support tooling that can flag not just *that* a patient is at elevated risk, but *which combination* of exposures is driving that risk and *what a reviewer could act on*.

Limitations & Future Work

This phase of the project was evaluated on two oncology exposure cohorts — the most complete DHS categories available in a comparatively sparse spontaneous-reporting dataset — so these results are an initial validation rather than a population-level claim, and inherit the general limitations of spontaneous-reporting data (missing information, reporting bias, no established causality). Neither ARISES nor MLX is inherently restricted to oncology or pharmacovigilance; extending the framework to larger and more diverse cohorts, additional pharmacovigilance databases, and adjacent problems — such as correlated complications in electronic health records, or interacting risk factors in chronic disease populations — is planned future work.

Project Status

- Manuscript prepared for submission to the IEEE Journal of Biomedical and Health Informatics, co-authored with Amruta Soma (equal contribution), Gandhali Sheth, Bo (Eric) Wang, Dr. Chun-Kit Ngan (WPI, corresponding author), and Dr. Yin-Ting Cheung (CUHK School of Pharmacy).
- A parallel academic report was prepared for the CUHK School of Pharmacy collaborators, and a research poster summarizing the project was produced for REU presentation.
- Both ARISES and MLX are implemented end-to-end, including the web-based prediction-and-explanation prototype described above.
- Full methodology, formulas, algorithms, and complete metric tables are withheld from this public write-up pending peer review and publication.

This document intentionally omits algorithmic specifics, formulas, and full metric tables to protect unpublished research prior to peer review. A complete technical treatment will be linked here once the associated manuscript is published.